

2 (a) e.g. initial speed is zero
 constant acceleration
 straight line motion
 (any two, one mark each)B2 [2]

(b) (i) $s = \frac{1}{2}at^2$
 $0.79 = \frac{1}{2} \times 9.8 \times t^2$ C1
 $t = 0.40$ s allow 1 SF or greater A1
 2 or 3 SF answer A1 [3]

(ii) distance travelled by end of time interval = 90 cm C1
 $0.90 = \frac{1}{2} \times 9.8 \times t^2$
 $t = 0.43$ s allow 2 SF or greater C1
 time interval = 0.03 s A1 [3]

(c) (air resistance) means ball's speed/acceleration is less M1
 length of image is shorter A1 [2]

3 (a) (i) frequency of light B1 [1]